

Experts Problem and Online Convex Optimization

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References

- Lecture 1 and 2 of [Haipeng's course](#)

Learning from Experts (The Experts Problem)

Suppose there are N experts (actions)

For $t = 1, 2, \dots, T$:

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss of every expert $\ell_t = (\ell_t(1), \dots, \ell_t(N)) \in \mathbb{R}^N$

Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle = \mathbb{E}_{i \sim p_t} [\ell_t(i)]$

A special case of Online Convex Optimization with

$\mathcal{X} = \Delta_N$ and $f_t(p) = \langle p, \ell_t \rangle$

$$\text{Regret} := \sum_{t=1}^T \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t=1}^T \ell_t(i)$$

Follow the Leader (FTL)

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Learner generates a distribution $p_t \in \Delta_N$

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Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

Define
$$L_t(i) := \sum_{s=1}^t \ell_s(i)$$

“Follow the leader” strategy: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$ and let $p_t = e_{i_t}$

Follow the Leader (FTL)

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

$$\text{Regret} = \underbrace{\sum_{t=1}^T \langle p_t, \ell_t \rangle}_{\geq T-1} - \underbrace{\min_{i \in [N]} \sum_{t=1}^T \ell_t(i)}_{\approx \frac{T}{2}} \approx (T-1) - \frac{T}{2} \approx \Theta\left(\frac{T}{2}\right)$$

Softmin

Replacing “min” by “softmin”:

$$p_t(i) = \frac{w_t(i)}{\sum_{j=1}^N w_t(j)} \quad \text{where} \quad w_t(i) = e^{-\eta L_{t-1}(i)} \quad \textbf{(Exponential weights)}$$

“Follow-the-leader” (hardmin) is a special case when $\eta = \infty$

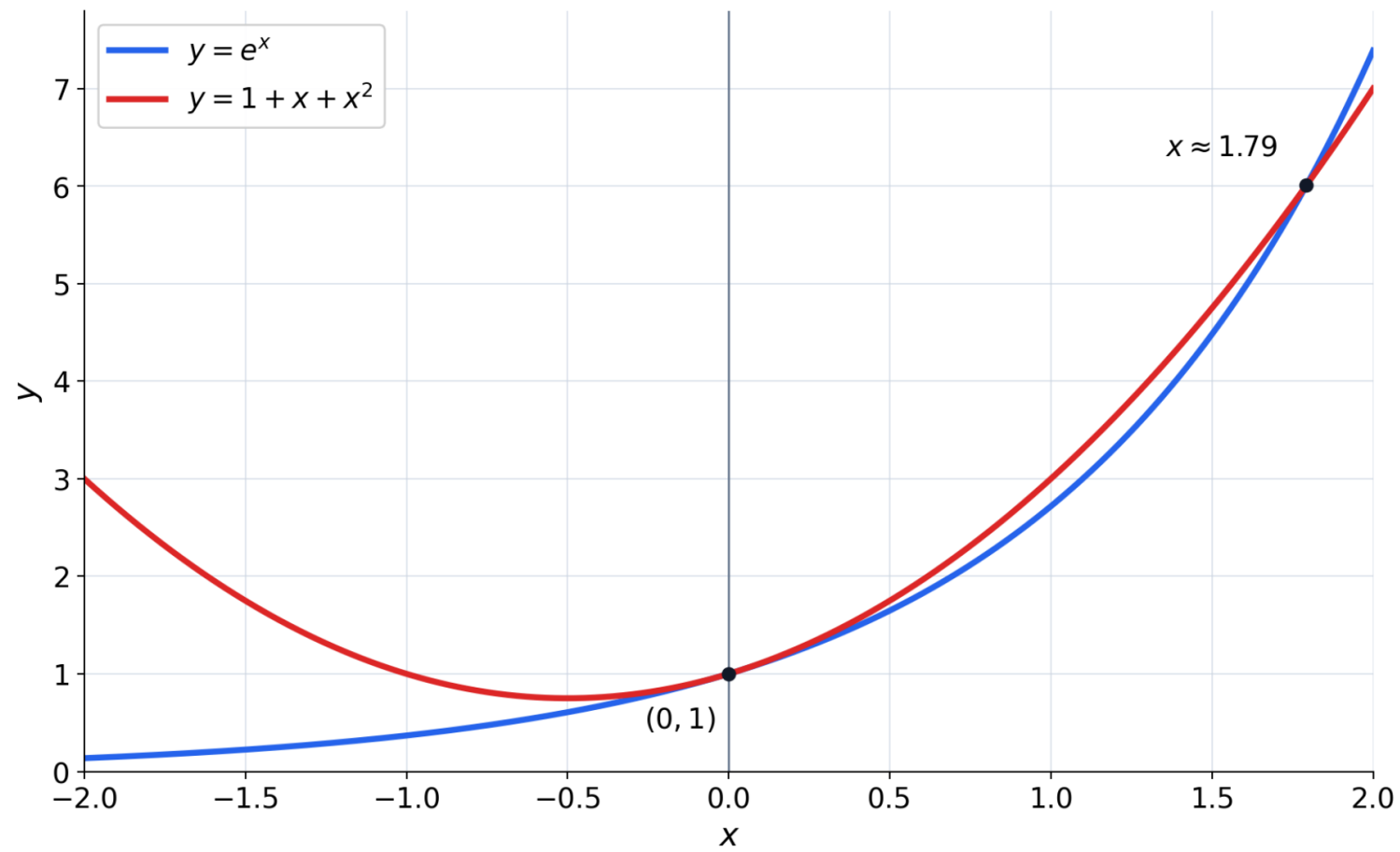
Regret Guarantee for Exponential Weights

Theorem.

Assume that $\eta \ell_t(i) \geq -1$ for all t, i . Then Exponential Weight ensures

$$\text{Regret} \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \sum_{i=1}^N p_t(i) \ell_t(i)^2$$

Regret Analysis for Exponential Weights

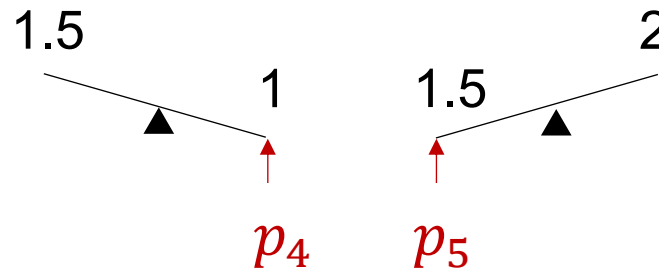
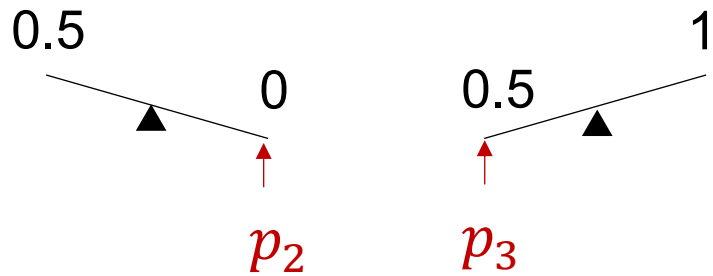


Why softmin works, while min does not?

The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

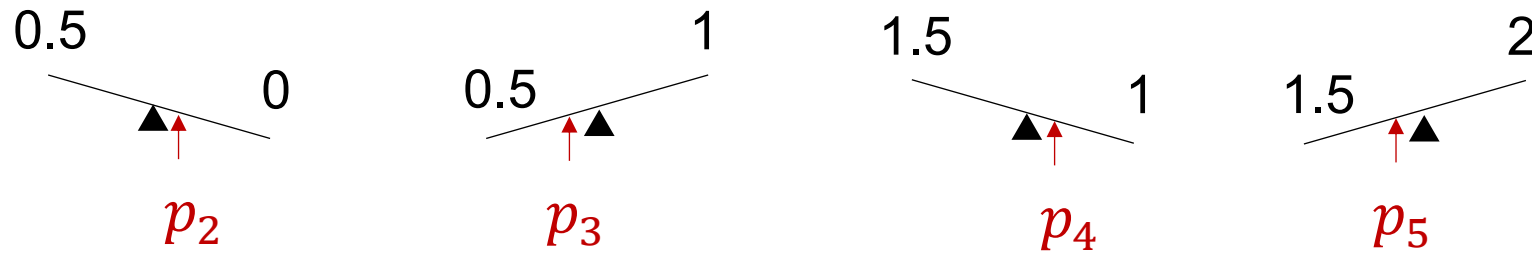
FTL: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$



The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

EW: $p_t(i) \propto e^{-\eta L_{t-1}(i)}$



Why Stability Helps?

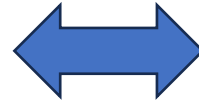
Follow-The-Regularized-Leader

(Generalizing the Exponential Weights algorithm)

An Equivalent Form of EW

$$\psi(p) := \sum_{i=1}^N p(i) \ln p(i) \quad (\text{convex})$$

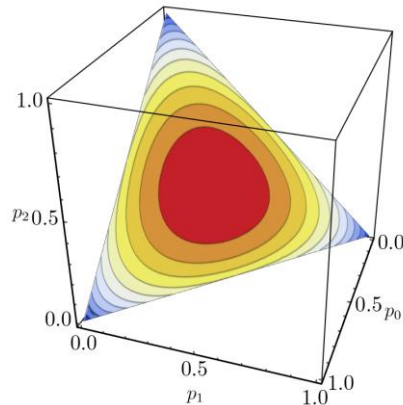
$$p_t(i) = \frac{e^{-\eta L_{t-1}(i)}}{\sum_{j=1}^N e^{-\eta L_{t-1}(j)}}$$



Verify yourself
(Lagrange multiplier)

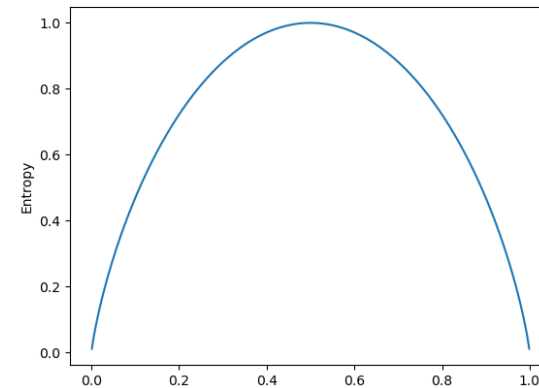
$$p_t = \operatorname{argmin}_{p \in \Delta_N} \left\{ \langle p, L_{t-1} \rangle + \frac{1}{\eta} \psi(p) \right\}$$

$-\psi(p) = \sum_{i=1}^N p(i) \ln \frac{1}{p(i)}$ is called the **(Shannon) entropy function** (concave)



N=3

[\(link\)](#)



N=2

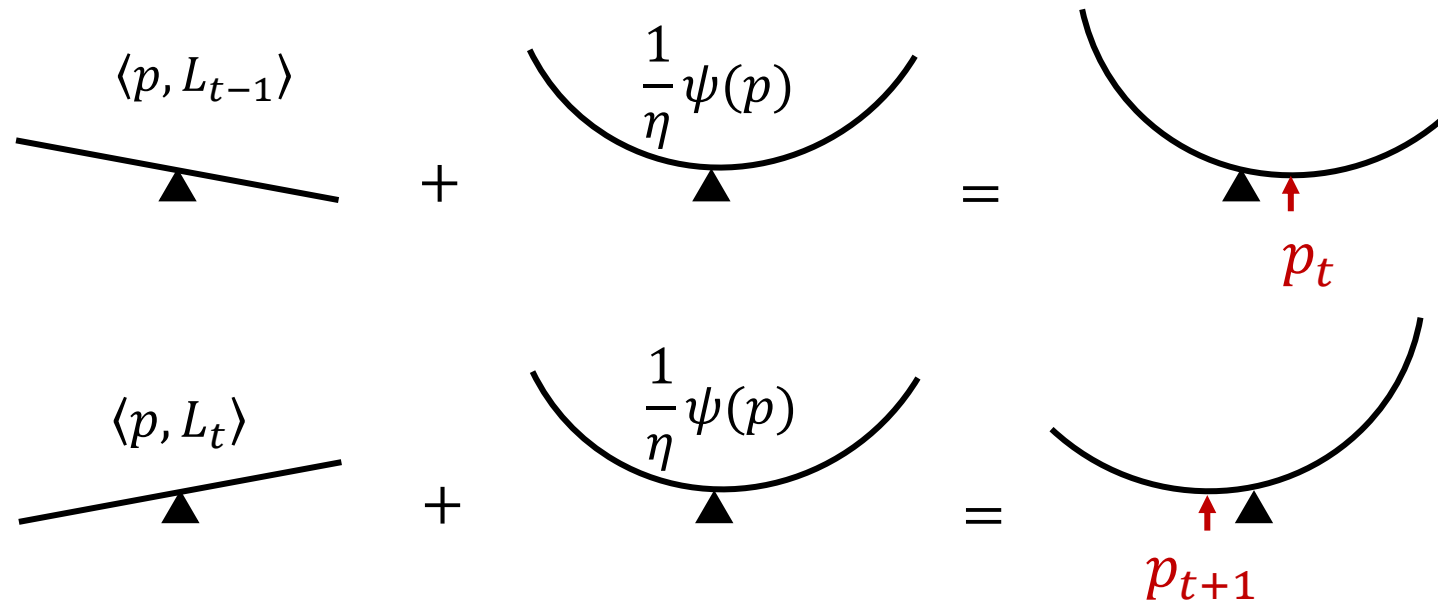
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Follow the Regularized Leader (FTRL)

$$p_t = \operatorname{argmin}_{p \in \Delta_N} \left\{ \langle p, L_{t-1} \rangle + \frac{1}{\eta} \psi(p) \right\}$$

Regularizer

If ψ is strongly convex, it makes the update *more stable*:



Strongly Convex Functions

FTRL for Online Linear Optimization

Online Linear Optimization (OLO)

Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For round $t = 1, 2, \dots, T$:

Learner makes a decision $x_t \in \mathcal{X}$

Adversary decides a loss vector $\ell_t \in \mathbb{R}^d$

Adversary reveals ℓ_t to the learner

FTRL

$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \langle x, L_{t-1} \rangle + \frac{1}{\eta} \psi(x) \right\}$$

A special case of Online Convex Optimization with $f_t(x) = \langle x, \ell_t \rangle$

$$\text{Regret} := \sum_{t=1}^T \langle x_t, \ell_t \rangle - \min_{x \in \mathcal{X}} \sum_{t=1}^T \langle x, \ell_t \rangle$$

Regret Guarantee of FTRL in OLO

Theorem. Let ψ be 1-strongly-convex in norm $\|\cdot\|$.

Then FTRL in OLO ensures

$$\text{Regret} \leq \frac{\max_{u \in \mathcal{X}} \psi(u) - \min_{u \in \mathcal{X}} \psi(u)}{\eta} + \eta \sum_{t=1}^T \|\ell_t\|_{\star}^2$$

$\|\cdot\|_{\star}$ is the dual norm of $\|\cdot\|$

Analysis (First-Order Optimality)

Analysis (Potential Evolution)

Analysis (Stability)